

FIELD THEORY PART-II

Phase 8: EM Wave Equation and Plane Wave

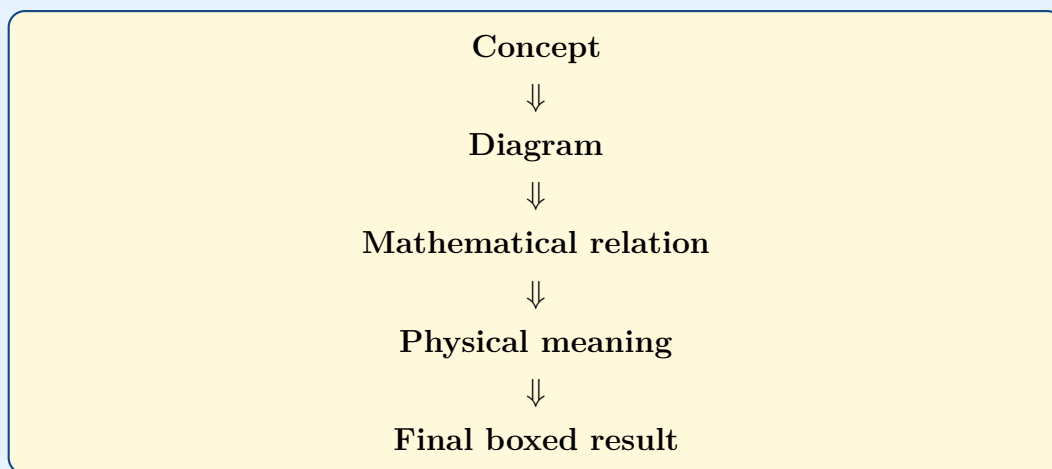
Maxwell equations to wave equation and analytical solution

Academic Article Style Notes for Semester Preparation

How to Use This Phase

This phase is made for semester preparation from a very basic level. Read the concept first, then understand the diagram, then learn the derivation pattern, and finally practise the exam-style writing block.

The recommended writing order is:



Contents

1	Aim of Phase 8	2
2	Maxwell Curl Equations	2
3	Wave Equation for Electric Field	2
4	Wave Equation for Magnetic Field	3
5	Free Space Wave Equation	3
6	Plane Wave Assumption	3
7	Analytical Solution	3
8	Exam Writing Template	4
9	Final Self-Test for This Phase	5

1 Aim of Phase 8

Main Goal

This phase prepares the most important long answer: electromagnetic wave equation and plane wave solution.

2 Maxwell Curl Equations

Starting Point

For a homogeneous conducting medium:

$$\mathbf{D} = \epsilon \mathbf{E}, \quad \mathbf{B} = \mu \mathbf{H}, \quad \mathbf{J} = \sigma \mathbf{E}.$$

Maxwell curl equations are:

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}$$

and:

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t}$$

3 Wave Equation for Electric Field

Derivation

Take curl of Faraday law:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}).$$

Substitute:

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t}.$$

Then:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu \sigma \frac{\partial \mathbf{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

Using:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}.$$

In source-free region:

$$\nabla \cdot \mathbf{E} = 0.$$

Therefore:

$$\nabla^2 \mathbf{E} = \mu \sigma \frac{\partial \mathbf{E}}{\partial t} + \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

4 Wave Equation for Magnetic Field

Final Equation

Similarly:

$$\nabla^2 \mathbf{H} = \mu\sigma \frac{\partial \mathbf{H}}{\partial t} + \mu\epsilon \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

5 Free Space Wave Equation

Free-Space Conditions

In free space:

$$\sigma = 0, \quad \mu = \mu_0, \quad \epsilon = \epsilon_0.$$

So:

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

and:

$$\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

6 Plane Wave Assumption

Wave Along +x

For a plane wave travelling along +x:

$$\frac{\partial}{\partial y} = 0, \quad \frac{\partial}{\partial z} = 0.$$

Take:

$$\mathbf{E} = E_y(x, t) \hat{\mathbf{y}}, \quad \mathbf{H} = H_z(x, t) \hat{\mathbf{z}}.$$

7 Analytical Solution

Electric Field Solution

$$E_y(x, t) = E_0 \cos(\omega t - \beta x)$$

For magnetic field:

$$H_z(x, t) = H_0 \cos(\omega t - \beta x)$$

where:

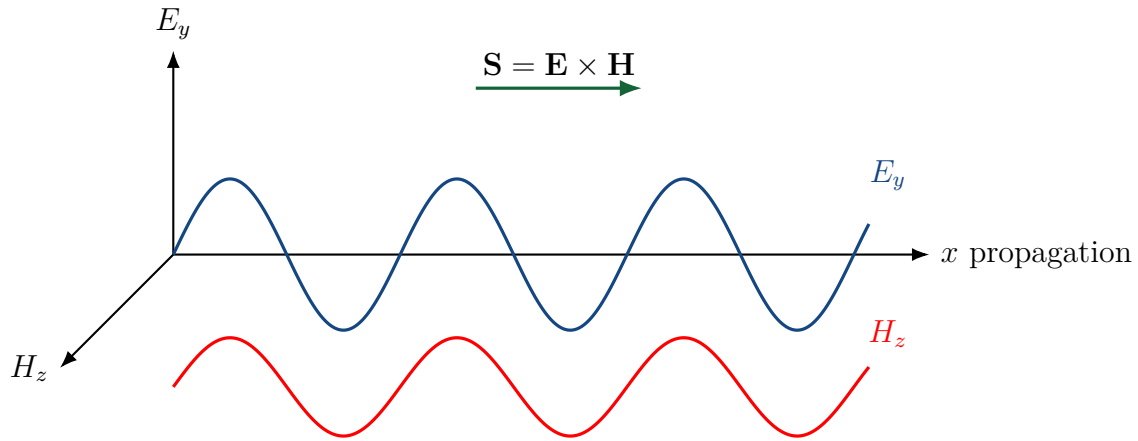
$$\beta = \omega \sqrt{\mu_0 \epsilon_0}$$

Velocity:

$$v = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

In free space:

$$c \approx 3 \times 10^8 \text{ m/s}$$



Plane EM wave: $\mathbf{E}$, $\mathbf{H}$ and propagation direction are mutually perpendicular

Figure 1: Plane electromagnetic wave propagation in free space.

8 Exam Writing Template

Long Answer Pattern

Write:

1. Maxwell curl equations.
2. Material relations.
3. Curl of curl identity.
4. Derive electric field wave equation.
5. Derive magnetic field wave equation.
6. Put free-space condition.
7. Plane wave assumption.
8. Sinusoidal solution and velocity.
9. Wave propagation diagram.

9 Final Self-Test for This Phase

Instructions

Answer these without looking at the notes. If you cannot answer a point, revise that section again before moving to the next phase.

1. Can I state the main definition of this phase?
2. Can I draw the main diagram neatly without overlap?
3. Can I explain the physical meaning in two or three lines?
4. Can I write the final formula with correct symbols?
5. Can I connect this topic with at least one past-year question pattern?
6. Can I write a semester-style answer with definition, diagram, derivation and final result?

Completion Rule

This phase is complete only when the concept, diagram, formula and answer-writing pattern are all ready.